

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS | GRADE 10 | SUB-STRAND 2.1: SIMILARITY AND ENLARGEMENT

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why does resizing a passport photo change its size but never its shape, and what is the mathematics that makes this possible? Use the Kimathi Street passport photo story (a 35 mm × 45 mm print and a 105 mm × 135 mm print of the same image) to answer every section. Write in full sentences, show every step of your working, and include units. Use the words from our lessons: linear scale factor (LSF), area scale factor (ASF), volume scale factor (VSF), corresponding sides, corresponding angles and similar. Write your answers in the space under each prompt.

Part 1: Finding the Number that Connects the Two Prints

A student in Nairobi has two prints of the same photo made at a studio along Kimathi Street. The small print is 35 mm wide and 45 mm tall. The large print for the school noticeboard is 105 mm wide and 135 mm tall.

(a) Calculate the ratio of the large width to the small width, and the ratio of the large height to the small height. Show your working.

(b) What do we call this number? Write the formula you used to find it.

(c) Is this an enlargement or a reduction? How do you know from the value of the number?

(d) The face on the small print has an eye-to-chin distance of 20 mm. Use the number from part (a) to find the eye-to-chin distance on the large print.

(a) Width ratio = $105 \div 35 = 3$. Height ratio = $135 \div 45 = 3$. Both ratios are equal to 3.

(b) This number is the Linear Scale Factor (LSF). $LSF = \text{length in the enlarged figure} \div \text{corresponding length in the original figure}$.

(c) It is an enlargement, because the LSF is 3, which is greater than 1. Every length on the large print is longer than the matching length on the small print.

(d) Every length is multiplied by the LSF, so the eye-to-chin distance on the large print is $20 \text{ mm} \times 3 = 60 \text{ mm}$.

Part 2: Why the Shape Is Kept, and Why a Distorted Photo Is Not Similar

The photographer accidentally makes a third print by tripling the width of the small photo but leaving the height at 45 mm. The face looks squashed and unrecognisable.

(a) Write the width and height of the distorted print and calculate the ratio of new width to old width and the ratio of new height to old height.

(b) State the two conditions that must both be true for two figures to be similar.

(c) Use your answer to (a) and the conditions in (b) to explain why the distorted print is not similar to the small print, while the 105 mm × 135 mm print is.

(d) Explain why similarity is a multiplicative relationship and not an additive one. Give one example from the photos.

(a) The distorted print is 105 mm wide and 45 mm tall. Width ratio = $105 \div 35 = 3$. Height ratio = $45 \div 45 = 1$. The two ratios are 3 and 1, so they are not equal.

(b) Two figures are similar if and only if (1) all corresponding angles are equal, AND (2) all ratios of corresponding sides are equal. This constant ratio is the LSF.

(c) For the distorted print the side ratios are 3 and 1, so condition (2) fails and there is no single LSF. The face is stretched sideways only, so it loses its shape. For the 105 mm × 135 mm print both ratios are 3, so every length is multiplied by the same LSF. The angles stay equal and the print is similar to the small one, which is why the face looks identical.

(d) Similar figures are linked by multiplying every length by the same number, not by adding the same amount to each length. For example, adding 70 mm to both sides of the small print would give 105 mm × 115 mm, where the ratios are 3 and about 2.56 and the shape changes. Multiplying both sides by 3 gives 105 mm × 135 mm and the shape is kept.

Part 3: From Lengths to Area and Volume

(a) Calculate the area of the small print and the area of the large print in mm². Then find the ratio of the large area to the small area.

(b) Using a small rectangle with sides l and w that is enlarged by LSF k , write an algebraic proof showing how the area changes. State the relationship between ASF and LSF.

(c) The photo had been a three-dimensional object instead of a flat print, and every length was multiplied by 3. State the relationship between VSF and LSF and calculate the VSF.

(a) Small area = $35 \times 45 = 1\,575$ mm². Large area = $105 \times 135 = 14\,175$ mm². Ratio = $14\,175 \div 1\,575 = 9$.

(b) The original area is $l \times w$. After enlargement the sides are kl and $k w$, so the new area = $(kl)(k w) = k^2(lw)$. The area is multiplied by k^2 , so $ASF = LSF^2$. For the photos, $ASF = 3^2 = 9$, which matches part (a).

(c) $VSF = LSF^3$. With $LSF = 3$, $VSF = 3^3 = 27$, so the volume would be 27 times larger.

(d) The student has only used the LSF, which describes lengths. Paper is measured by area, so we must use $ASF = LSF^2 = 9$. The large print uses 9 times as much paper, not 3 times.

(d) A student says: 'The large print is 3 times as big, so it uses 3 times as much paper.' Explain what is wrong with this statement.

Part 4: Choosing the Correct Scale Factor in Kenyan Problems

Before you calculate, state which scale factor (LSF, ASF or VSF) the problem gives you and which one you need.

(a) A maize plot in Kericho is represented by a plan. The real plot is similar to the plan, with LSF = 5 from plan to real plot. The plan has an area of 120 m². Find the area of the real plot.

(b) A road sign near Nakuru is similar to a model sign. The model sign has an area of 900 cm² and the real sign has an LSF of 4 compared with the model. Find the area of the real sign.

(c) A small water tank on a Kericho tea farm holds 40 litres. A similar larger tank has LSF = 3 compared with the small one. Find the capacity of the larger tank.

(d) A map of Nairobi to Nakuru uses a scale of 1 : 500 000. The distance between the towns on the map is 18 cm. Find the real distance in kilometres.

(a) Given: LSF = 5. Needed: ASF, because the problem is about area. $ASF = 5^2 = 25$. Real area = $120 \times 25 = 3\,000\text{ m}^2$.

(b) Given: LSF = 4. Needed: ASF. $ASF = 4^2 = 16$. Real area = $900 \times 16 = 14\,400\text{ cm}^2$.

(c) Given: LSF = 3. Needed: VSF, because capacity is a volume. $VSF = 3^3 = 27$. Larger capacity = $40 \times 27 = 1\,080$ litres.

(d) Given: LSF = 500 000 (map to real). Needed: LSF, because the question is about a length. Real distance = $18 \times 500\,000 = 9\,000\,000\text{ cm}$. Converting, $9\,000\,000\text{ cm} \div 100\,000 = 90\text{ km}$.

Part 5: The Final Explanation

Write a complete answer to the driving question: Why does resizing a passport photo change its size but never its shape, and what is the

Resizing a passport photo changes its size but not its shape because the photographer's software multiplies every length by the same number, the Linear Scale Factor. For the 35 mm × 45 mm print and the 105 mm × 135 mm print, $105 \div 35 = 3$ and $135 \div 45 = 3$, so LSF = 3. When every length is multiplied by the same LSF, all corresponding angles stay equal and all ratios of corresponding sides stay equal. These are the two conditions for similar figures, so the two prints are similar and the face looks identical. In the distorted print only the width was multiplied by 3 and the height was multiplied by 1. The side ratios (3 and 1) are

mathematics that makes this possible? Your answer must: – refer to the 35 mm × 45 mm and 105 mm × 135 mm prints and the distorted print; – explain the role of the LSF, corresponding angles and the ratios of corresponding sides; – explain what happens to area and to volume; – explain what the centre of enlargement tells us about how the photo was resized.	not equal, so the figures are not similar and the face is squashed. The LSF also controls area and volume. Areas change by $ASF = LSF^2 = 3^2 = 9$, which we confirmed with 1 575 mm² and 14 175 mm², so the large print uses 9 times as much paper. For a three-dimensional object, volumes would change by $VSF = LSF^3 = 27$. The resizing works from a centre of enlargement. Lines drawn through each pair of corresponding points on the small and large prints meet at one point, and every point of the image is 3 times as far from this centre as the matching point on the original. This is like the camera lens position. Because one LSF is applied from one centre to every point, the shape is perfectly preserved while the size changes.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Calculating and interpreting the Linear Scale Factor	Correctly calculates $LSF = 3$ from both dimensions, writes the formula, identifies enlargement because $LSF > 1$, and uses it accurately to find unknown lengths, with units and clear working.	Calculates $LSF = 3$ and uses it to find a missing length, but the formula, the enlargement reasoning or the units are incomplete.	Calculates only one ratio or makes arithmetic errors, or cannot link the number to the term Linear Scale Factor or use it to find lengths.
Explaining similarity using the two conditions	States both conditions (equal corresponding angles and equal ratios of corresponding sides) and uses the ratios 3 and 1 of the distorted print to justify clearly why it is not similar, including why similarity is multiplicative.	States both conditions and identifies that the distorted print is not similar, but the use of ratio values or the multiplicative idea is thin.	States only one condition or says the distorted print is not similar without evidence from ratios.
Relating LSF to ASF and VSF	Calculates both areas, finds the ratio 9, proves $ASF = LSF^2$ algebraically, states $VSF = LSF^3 = 27$, and clearly corrects the 'three times the paper' misconception.	Gets $ASF = 9$ and $VSF = 27$ correctly, but the algebraic proof or the misconception explanation is incomplete.	Uses LSF in place of ASF or VSF, or cannot show why the area changes by the square of the LSF.
Choosing and applying the correct scale factor in Kenyan contexts	Names the scale factor given and needed for each problem, then applies LSF, ASF or VSF correctly with accurate calculations and unit conversions.	Chooses the correct scale factor for most problems and calculates correctly, with a small slip in a unit or a missing diagnosis statement.	Uses the wrong scale factor in several problems, for example multiplying an area by the LSF.
Final Explanation of the driving question	Gives a coherent, connected explanation that uses the passport photo data, LSF,	Explains the role of LSF and the similarity conditions with the photo data, but area,	Gives a general description of resizing without referring to the LSF, the data or

	both similarity conditions, ASF, VSF and the centre of enlargement, with accurate mathematical vocabulary.	volume or the centre of enlargement is missing or weakly linked.	the similarity conditions.
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